

Are the Molecules That Make a Solution Red Big or Small?

A teaching experiment produced by the

POGIL-PCL Group (2015)
Version 1.0.1

Lead Author
Robert Whitnell, Guilford College

We gratefully acknowledge support from The National Science Foundation (DUE-1044624) and the POGIL Project. We would also like to recognize the contributions of the many members of the POGIL-PCL group who reviewed, tested, and helped to produce this experiment in final form.

This work is licensed under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License. To view a copy of this license, visit <http://creativecommons.org/licenses/by-nc-sa/4.0/> or send a letter to Creative Commons, PO Box 1866, Mountain View, CA 94042, USA.



Instructor's Handbook

Are the Molecules That Make a Solution Red Big or Small?

Versions:

1.0.1 A couple of questions were altered to remove some of the circular reasoning. This is a minor change from the previous version.

Unlabeled This is the first version that was released through the POGIL-PCL Google Drive.

Learning Objectives: Content

Students will

- correctly describe that a spectrum's y-axis relates to molecular concentration and x-axis relates to structure
- determine molar absorptivity graphically and determine the concentration of an unknown spectroscopically
- apply Beer's Law to a mixture of absorbing molecules to understand their independent behavior in a dilute solution
- relate particle-in-a-box theory to spectroscopy

Learning Objectives: Process

Students will

- model data with a linear relation
- make solutions, including serial dilutions, using volumetric glassware and micropipettes
- describe and interpret the data obtained from a UV-Vis spectrophotometer

This experiment investigates the UV-Vis absorption spectra of one or more series of conjugated cyanine dyes. There are multiple variations of cyanine dyes, however only one series is presented here.

Timing/Number of Lab Periods Required

In most cases, the full lab can be completed in two four-hour or three three-hour lab periods, assuming that students arrive at the first lab with the pre-lab questions completed and ready to discuss. For four-hour labs, the natural breaking point is after Experiment Part Two, including discussion of the Thinking about the Data questions. For three-hour lab periods, a reasonable first breaking point is either after solution preparation or data collection for Experiment Part Two, and the second breaking point is after Experiment Part Three, including discussion of the Thinking about the Data questions. Instructors who choose to skip Experiments Part One and/or Part Two (see the discussion below) should be able to complete this experiment in two lab periods.

Alternate versions for approaching Beer's Law

There are two approaches to Part One of this experiment, which addresses Beer's Law considerations. The first approach focuses on having students uncover Beer's Law by observing how concentration affects absorbance spectra. Students determine the extinction coefficient for each dye. The other approach assumes students know Beer's Law and focuses on making solutions of desired absorbance using Beer's Law considerations. Because the solutions are unlabeled, students in both approaches match the solution to the correct molecule.

Materials/Equipment

Any series of cyanine dyes will work for this experiment. Here are three series that are available from Sigma-Aldrich. Other series, such as the diphenyl dyes, are also possible.¹

3,3'-thiacyanine series	CAS number
3,3'-diethylthiacyanine iodide	[2197-01-5]
3,3'-diethylthiacarbocyanine iodide	[905-97-5]
3,3'-diethyldicarbothiacyanine iodide	[514-73-8]
3,3'-diethyltricarbothiacyanine iodide	[3071-70-3]

4,4'-carbocyanine series	CAS number
1,1'-diethyl-4,4'-cyanine iodide	[4727-49-5]
1,1'-diethyl-4,4'-carbocyanine iodide	[4727-50-8]
1,1'-diethyl-4,4'-dicarbocyanine iodide	[18300-31-7]

2,2'-carbocyanine series	CAS number
1,1'-diethyl-2,2'-cyanine iodide	[977-96-8]
1,1'-diethyl-2,2'-carbocyanine iodide	[605-91-4]
1,1'-diethyl-2,2'-dicarbocyanine iodide	[14187-31-6]

25 mL of a 0.001M solution of each solution will be sufficient stock solution, even for the largest classes. The solutions should not be labeled with the name of the dye; solutions A, B, C, and D should be sufficient. Of course, the instructor knows which dye is in each solution.

A UV-Vis spectrometer that scans from 400-850 nm. The cuvettes should be impervious to methanol.

¹ Anderson, J. J. *Chem. Educ.*, **1997**, 74, 985.

Students should have access to computers that can download and run PhET software (<https://phet.colorado.edu>). In order to do all of Part Four, students should have access to an electronic structure program, such Gaussian, Hyperchem, Spartan, etc.

Are the Molecules That Make a Solution Red Big or Small?

This experiment investigates the UV-Vis absorption spectra of a series of the conjugated cyanine dyes.

Spectra contain useful information about both the number of molecules present and the structure of the molecules. We often use data from the y-axis of the spectrum to monitor the concentration of some molecule as a function of time. In this experiment we will use both y-axis and x-axis spectral data for the UV-Vis spectrum in a study of a series of cyanine dyes.

Pre-experiment questions: Part One

1. Write a definition of λ_{max} .

The definition of λ_{max} should be something like “the wavelength in the spectrum at which the absorbance is a maximum.”

2. A certain molecule has a strong absorbance spectrum with a λ_{max} of 500 nm when dissolved in water.
 - a. Sketch a spectrum for this molecule in your lab notebook, including x-axis and y-axis numbers and labels.

Students should sketch a broad peak centered at 500 nm. Be sure that students include proper axis labels.

- b. Draw a second spectrum on the same x-y axes for a solution that has the concentration of the molecule decreased by a factor of 2.

Students will sketch a broad peak half as intense as the spectrum from question 1.

3. To do in class: Compare your answers to questions 1 and 2 to those of your group, and select the correctly drawn spectra. Draw these spectra on the board.

Most students have very similar answers for questions 1 and 2. Class discussion can quickly address any questions or misconceptions.

4. Describe Beer's Law. Label all the variables (include units) in the Beer's Law equation.

$A = \epsilon bc$ where ϵ is the molar absorptivity ($1/(\text{M cm})$), b is the path length (cm) and c is the concentration (M).

5. Show the calculations needed to prepare a 0.001 M solution of a solid compound with a molar mass of $466.40 \text{ g mol}^{-1}$. Use methanol as the solvent. Given the equipment available, can the calculated mass of the compound be measured precisely? Describe how the solution should be prepared. Include all glassware needed to make the solution.

Students quickly realize a key piece of information missing in this question and the next one: the volume of the solution. They will often choose a volume that is too small (such as the amount needed to fill a cuvette) or much too large (they don't need a 1 L solution). Class discussion can focus on appropriate volumes for this solution. If the calculated mass is too small for the available balances, discussion can also address making a smaller volume of a more

concentrated solution and then diluting that solution to the desired concentration. The description of how to make the solution should refer to balanced of the desired precision and proper use of volumetric glassware. For a 25 mL volumetric flask, one needs 11.7 mg of material.

6. Assume the compound from the previous question has a molar extinction coefficient (molar absorptivity) of $\sim 5 \times 10^5 \text{ L mol}^{-1} \text{ cm}^{-1}$. You want to prepare a solution that has an absorbance of 0.5. What are the necessary calculations to prepare it from the 0.001M stock solutions? Describe how to make the solution, including all glassware.

This question refers purely to dilution. The discussion in the previous question otherwise applies.

7. Look up *cyanine* in Wikipedia. List at least four important distinguishing characteristics specific to the cyanine structure (e.g., functional groups, double/single bonds, geometry, heteroatoms).

Students should identify the presence of 1) alternating double bonds, 2) planarity, 3) aromatic or conjugated systems, and 4) heteroatom(s).

8. Look up the structure of 1,1'-diethyl-4,4'-cyanine iodide by going to www.sigma-aldrich.com and searching the CAS number [4727-49-5]. Sketch the structure of this molecule in your lab notebook. Identify any important features in your drawn structure listed previously for the cyanine dyes. In addition, identify any features specific to this cyanine.

9. Look up the structures of 1,1'-diethyl-4,4'-carbocyanine iodide and 1,1'-diethyl-4,4'-dicarbocyanine (CAS numbers [4727-50-8], [18300-31-7]), and sketch their structures in your notebook. What structural feature changes in this series of cyanine dyes? Be specific.

Substitute the names of molecules to match the series of cyanines used in your lab. For each molecule, provide the CAS numbers and have the students look up the structures.

Students should identify a change in length of polymethine chain.

10. Review the MSDS for each dye in questions 8 and 9 and for methanol, the solvent used in this experiment. Write in your lab notebook any considerations for the safe use of these chemicals.

11. You have a solution of a molecule that is green. Predict the value of λ_{max} for that molecule.

Students will give a variety of answers. Some students will think that λ_{max} is the color of the solution; others will correctly remember that the solution color represents the transmitted light, not the absorbed light. Take note of these predictions, but the detailed discussion can be saved for after students have obtained the spectra.

12. The λ_{max} for 1,1'-diethyl-4,4'-cyanine iodide is 592 nm. Roughly sketch the UV-Vis spectrum of this molecule in your lab notebook. Then add, on the same axes, a predicted spectrum for each of the next two dyes in the series (carbo and dicarbo). Label your graph.

Given one value for λ_{max} , have students discuss in groups the positions of the absorbance peaks of other dyes in the series. This is accomplished without information or prompting of the relationship between conjugated chain length and absorbance wavelength. It may be more beneficial to use different dye molecules here since Sigma-Aldrich provides λ_{max} values for the

molecules that have been looked up in steps 2 and 3. Even if students know the values for all three dyes, discussion can still focus on why the trend is present.

13. To do in class: Compare your predicted spectra to those of your group; as a group, decide on one common answer and draw this on the board. Then predict the color of each dye solution. As a group, justify your predictions using complete sentences. You will read these out loud to the class.

Have a group member draw predicted spectra on the board. Discuss the validity of each plot.

Experimental Protocol

- A. Prepare the needed solutions. You will be provided with solutions of a known concentration in units of g/100 mL or ~0.001 M.

Monitoring student preparation of the solutions is valuable. While they will still get reasonable results even if they use graduated cylinders for measuring volume, this step represents a good opportunity to emphasize proper lab technique.

- B. Set up the spectrometers to record absorbance vs. wavelength.

Instructions to be provided by the instructor.

- C. Calibrate the spectrometers using methanol. Be sure to use cuvettes insensitive to organic solvents!

Instructions to be provided by the instructor.

- D. Decide what wavelength range is appropriate to record for the spectrum of the compound. Using the same cuvette, record the UV-Vis spectrum of the compound. Save the spectrum, and record λ_{max} .

Have students scan over at least the range 400-850 nm. Instructors can direct students to consult with them on the wavelength range before they take spectra.

Experiment Part One (Focus on review of Beer's Law)

Focus question: How does changing the concentration of a solution change its UV-Vis spectrum?

Each section starts with a focus question to motivate the experimental work. Students are not expected to answer this question prior to doing the experiment. Instructors are encouraged to use the answers to focus questions as student predictions.

For Parts One and Two, instructors may feel that their students have addressed the concepts sufficiently in previous physical chemistry experiments or previous classes. It may then be reasonable to start with Experiment Part Three, but the focus questions for Parts One and Two could then be used as diagnostic questions for how well students recall and understand the Beer's Law concepts.

Groups are expected to use Beer's Law to determine appropriate solution dilutions. Unless micropipets are available the dilution needs to be performed in two steps since the aliquot volume required is very small.

While this part of the experiment could be done with only one of the dyes, students typically decide to collect data on all three dyes, dividing the work among the class.

Each set of data should be saved to be transferred to a common location and available for the entire class.

You are provided with prepared stock solutions of the cyanine molecules in methanol with a concentration of $\sim 0.001\text{ M}$. The extinction coefficient for a typical cyanine is *about* $(1-7) \times 10^5\text{ M}^{-1}\text{cm}^{-1}$. Decide on the range of concentrations needed to determine the effect of concentration on the spectrum, and make the needed dilutions. Aim for absorbances between about 0.2 and 0.8. (Based on characteristics of the instrument, consider why this range is a useful one.) You may need to do additional dilutions in order to meet this target absorbance value.

Obtain spectra of the dye series over the chosen range of concentrations. Upload the spectra for access by the entire class.

Thinking About the Data: Part One

14. Plot the spectra collected by the class on the same axes. In what ways are the spectra the same? In what ways do the spectra differ?
15. Based on the class prediction in pre-experiment question 13, state which dye is present in each solution.
16. Calculate the concentration of each of your dye solutions in units of molarity. Decide how concentration qualitatively affects the UV-Vis spectrum of a molecule.

Students should now be able to identify the solution with the largest λ_{max} as having the dye with the longest conjugated chain.

17. Recall Beer's Law from the pre-experiment questions. Develop a method that uses the spectra of different concentrations of dye to determine graphically the dye's extinction coefficient, identifying the independent and dependent variables and the slope and intercept.

Absorbance should appear to be roughly proportional to concentration.

18. Use the method developed in question 0 to obtain appropriate spectra of the dye solutions. Plot the graph in Excel and determine the extinction coefficient for each dye *with uncertainty*. You may need to adjust your spectra baselines. Also, you may need to calculate an average value of the absorption at λ_{max} (consider the noise at the peak).

Students should use the method they develop to construct a linear plot of $\mathcal{A}$ (dependent variable) vs. c (independent variable) for each dye.

19. Describe any trends you observe in the extinction coefficients. (The post-experiment questions address the extinction coefficient in greater detail using additional information gathered in the remaining parts of this experiment.)

Students should see that the dye with the longest chain has the largest extinction coefficient. They will explore that more in the post-experiment questions at the end of the lab.

Experiment Part One (Focus on application of Beer's Law)

Focus question: How can Beer's law be used to determine the concentration of the dye solution?

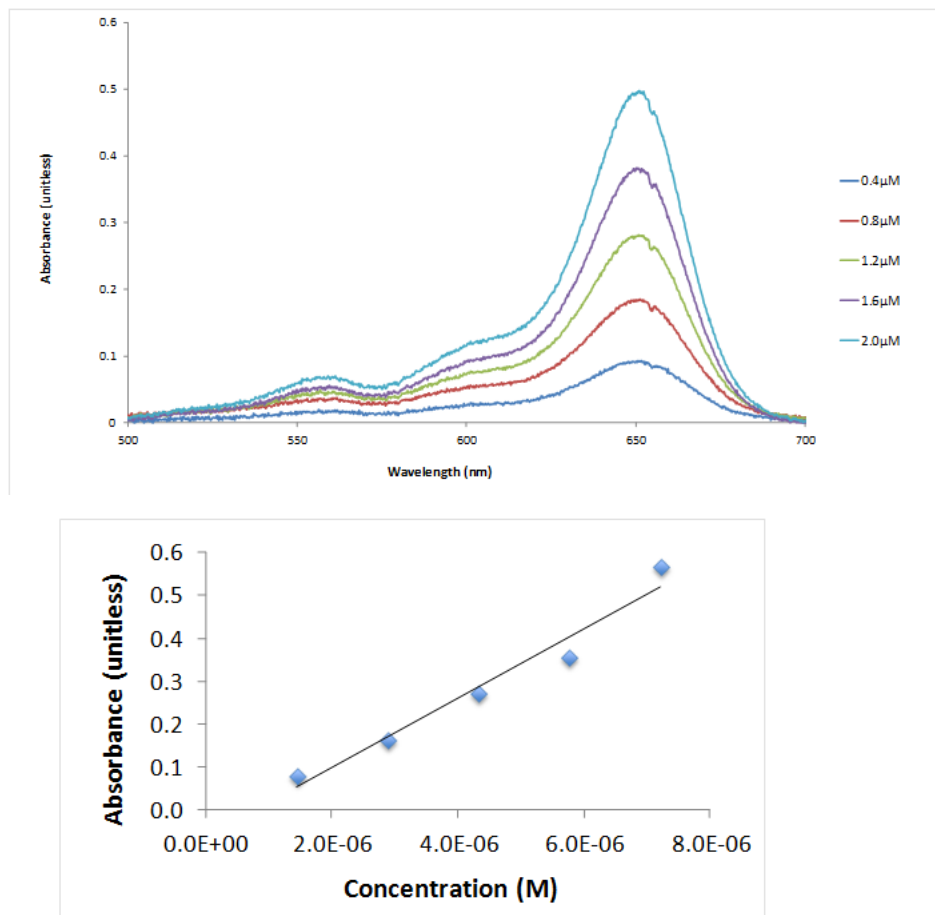
The comments on the other version of Part One apply here.

You are provided with prepared stock solutions of the cyanine molecules in methanol with a concentration of ~ 0.001 M. The extinction coefficient for a typical cyanine is *on the order of* $5 \times 10^5 \text{ M}^{-1} \text{ cm}^{-1}$. Do the calculations needed determine how to dilute the stock solution so that it has an absorbance between about 0.2 and 0.8. (Based on characteristics of the instrument, consider why this range is a useful one.) You may need to do multiple dilutions in order to meet this target absorbance value.

Obtain spectra of the dye series over the chosen range of concentrations. Upload the spectra for access by the entire class.

Thinking About the Data: Part One

14. Summarize any serial dilutions made to produce the solution that gave a spectrum with an absorbance between 0.2 and 0.8, explaining how you used Beer's Law in your calculations.
15. What is the concentration in g/100 mL of the dye in solution for which you recorded a spectrum?
16. Is Beer's Law dependent on the x-axis or y-axis of a spectrum? Discuss.
17. Plot spectra of the all three dyes on the same axes as three separate series.
18. Based on the class prediction in pre-experiment question 12, state which dye is present in each solution. Does your answer depend on the x-axis or y-axis of the spectra? Discuss.
19. Describe any trends you observe in the extinction coefficients. (Post-experiment question 6 addresses the extinction coefficient in greater detail using additional information gathered in the remaining parts of this experiment.)



Spectra and Beer's Law plot for a dicarbocyanine dye.

Pre-experiment Questions: Part Two

20. Sketch a predicted spectrum of two of the dyes mixed together. In doing so, select concentrations of the dye that give an absorbance at λ_{max} , for each individual dye, of between 0.2 and 0.8.

Students will typically predict that the spectrum will have two peaks (the sum of the spectra for the two dyes) or one peak at a value of λ_{max} between the values for the individual dyes (the average of the spectra for the two dyes).

21. Show the predicted spectrum to your instructor or draw it on the board.

Note any differences between predictions in the class discussion, but do not attempt to reach a resolution—that will be done once experimental data is available.

Experiment Part Two

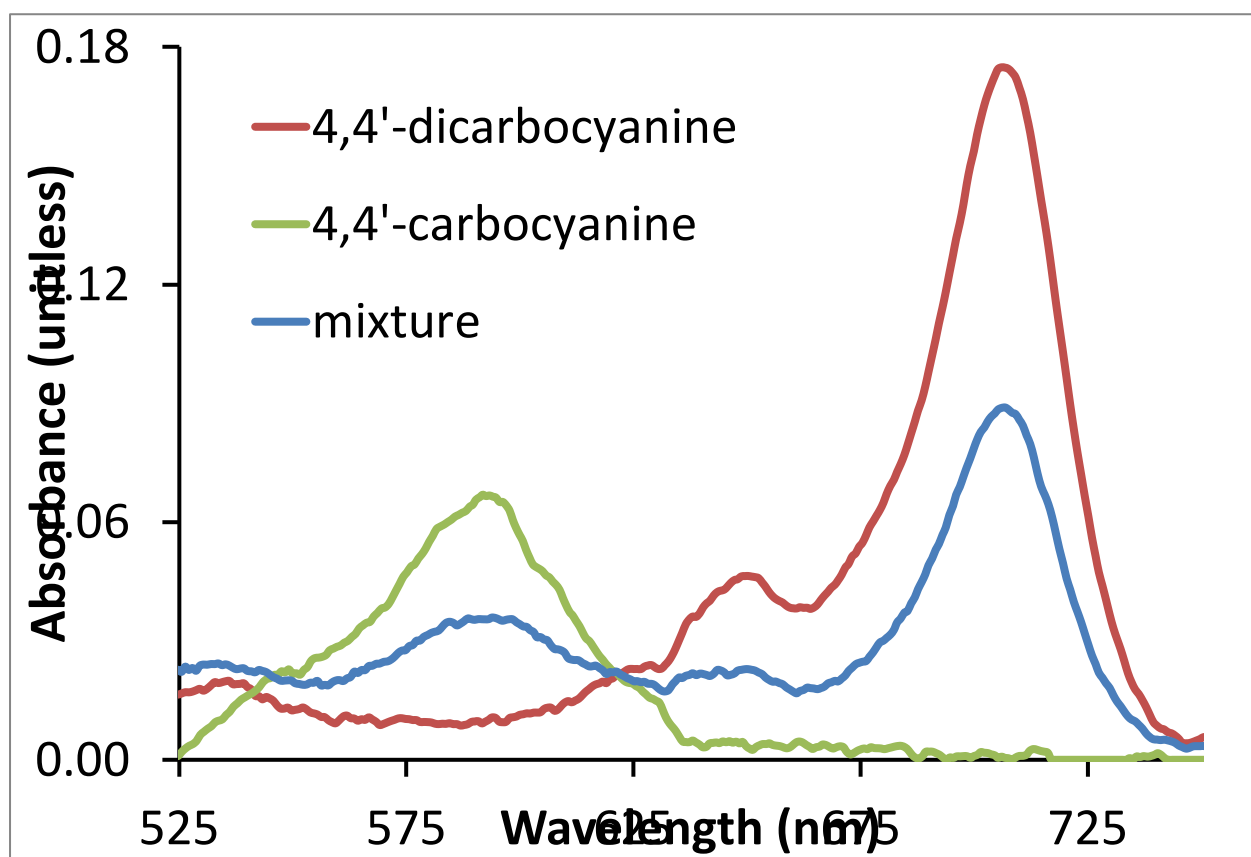
Focus question: How does mixing two dyes affect the UV-Vis spectrum?

As noted above, if instructors believe that students are sufficiently comfortable with this concept, this part could be skipped. The focus question can then be used as a diagnostic to gauge student understanding.

The class should divide up the work obtaining the spectra. If the class is large enough, it can be valuable to obtain duplicate spectra for some mixtures as a check on reproducibility. Spectra will be compared in front of the class after spectra are obtained.

Obtain spectra of dye mixtures, and upload the spectra for access by the entire class. As a class, decide which mixtures to use in order to best determine the effect of mixing.

Here's an example of a mixture spectra for three cyanine dyes.



Thinking About the Data: Part Two

22. Plot the spectra of the mixed dyes, and compare them to the spectra of the individual dyes. In what ways are the spectra the same? In what ways do the spectra differ?

Students should start to see here that the spectrum of the mixture is the sum of the spectra for the two dyes.

23. Does Beer's Law apply to mixtures of dyes (or to any mixtures)? Justify your answer using your data.

Some students see intuitively that Beer's Law still applies. Other students need to write out the equations in detail. They may need to be prompted that each dye should make an independent contribution and that they diluted the dye solutions in making the mixture.

24. How and why can UV-Vis spectroscopy be used to identify molecules in mixtures? What are the limitations of using UV-Vis spectroscopy to identify molecules?

Student discussion can focus on how identification may be possible if the molecules in the mixture are known to have different values of λ_{max} .

Pre-experiment Questions: Part Three

25. For the largest of the dye molecules, consider the shortest chain of carbons that connects the two nitrogen atoms. Draw the resonance structures for this chain. Are the carbon-carbon bonds in the chain identical, or are some single bonds and some double bonds? Are the two carbon-nitrogen bonds identical? Can this chain be considered to be a conjugated chain?

Students should develop a sense that all C-C bonds are equivalent as are the two C-N bonds at the end of the chain. The chain is, therefore, conjugated. If you choose to use a dye series that does not have a carbon-nitrogen bond, modify the above question appropriately.

Experiment Part Three

Focus question: How does molecular structure affect the UV-Vis spectrum?

Following the general protocol, obtain spectra for the cyanine series of dyes described in the pre-experiment questions.² Once each group has recorded its spectra, the saved file with the spectra should be named according to the dye and uploaded for access by the entire class.

If students obtained good spectra for all dyes in Part I, then the instructor can choose to have the students use those spectra. Only one spectrum for each dye is needed because the focus of this part of the experiment is on λ_{max} .

Thinking About the Data: Part Three

26. Plot the spectra for the different dyes on one graph, selecting spectra with intermediate absorbance values. Tabulate all λ_{max} values.

Compile spectra in a format to display to the entire class. Beginning a table with dye and λ_{max} values is helpful at this point. Data will be added to this table as students work through these questions.

27. Compare your experimentally determined values of λ_{max} to your predicted values from the pre-experiment questions for Part One. Do your results agree with your predicted trend?

² Consult with your instructor to determine whether the spectra obtained in Part I are suitable for this analysis.

If they haven't before now, students should now establish the trend of the longer conjugated chain correlating with larger values of $\lambda_{\max}$.

28. Current models of molecular structure postulate that the electrons in a molecule are confined to *molecular orbitals*, which are similar to atomic orbitals but “belong” to the molecule. Such orbitals can be localized over one or two atoms or can span a much larger number of atoms. UV-Visible spectra can be thought of as measuring the energy needed to excite an electron to a higher energy molecular orbital. For these dye molecules, some models of how much energy is needed rely on an effective length of the orbital. For each dye, estimate molecular orbital lengths based on each of the following assumptions:
- The orbital can span the entire molecule. Therefore, estimate the largest end-to-end distance in the molecule.
 - The orbital is bounded by the two ends of the conjugated chain. Estimate the length of this conjugated chain.

These two estimates provide a range of the size of the molecular orbital.

Students may have several ways of estimating the orbital size. They can count bonds and use an average bond length to determine the size. They can also build the molecule in a program such as Avogadro and use tools in those programs to measure the distances. These instructions are not so prescriptive that all students will estimate the orbital size in the two cases exactly the same way. That's o.k. if the class is sufficiently large to provide a good range of estimates and if students can clearly explain how they made their estimates. Instructors should encourage discussion of these estimates among different groups. For a smaller class, the instructor may wish to guide students to appropriate estimates.

29. One such model of molecular structure (the particle-on-a-line model of conjugated systems) leads to the following equation:

$$\lambda_{\max} = \frac{8mcL^2}{h(N+1)}$$

where m is the mass of the electron, c is the speed of light, and h is Planck's constant. L may be thought of as the effective length of the orbital to which the delocalized electrons are confined. N is the total number of delocalized electrons.

- Determine the number of delocalized electrons for each dye by considering the conjugated carbon atoms in the carbon chain and including the delocalized electrons from the attached nitrogen atoms. A nitrogen atom without a lone pair contributes one electron, and a nitrogen with a lone pair of electrons contributes two electrons.

N can be determined via the formula $N = \# \text{ carbon atoms} + 3$; the nitrogen with a double bond has one electron and the other a lone pair of electrons. The length of the molecular orbital will extend beyond the nitrogen atoms (to the saturated carbon connected to the nitrogen), but that will be taken into account later. Students may need some help to understand how many electrons are delocalized.

- b. View the spectra of the dyes to determine λ_{max} . Calculate the length, L , in angstroms and nm for each dye using the values of λ_{max} for each dye. Make a table with λ_{max} , N , and L values.

You may want to prompt your students to use a spreadsheet for these calculations. That is valuable in the later questions. The major issue in these calculations is likely to be units. Once they get the units right, they should start seeing L values that are near the values they estimated.

30. Which estimate from question 28 gives better agreement with the values of L calculated from the model in question 29? What does this result imply about the spatial extent of the electrons in the molecular orbitals responsible for this spectral transition?

Students should reach the conclusion that the spatial extent of the molecular orbital is primarily determined by the conjugated chain even though there is further conjugation in the rings.

31. The effective length of the orbital, L , can in turn be related to the experimentally determined average C-C bond length, l , along the conjugated carbon chain according to the equation³

$$L = (b \times l) + \gamma$$

where b is the number of bonds in the chain, and γ is a property of the dye that depends on the functional groups attached to the nitrogen atoms at each end of the chain.

- a. Examine this equation, and decide which variable is the independent and which is the dependent variable. Identify the slope and intercept.

γ accounts for the extension of the molecular orbital beyond the nitrogen atoms or whatever is the appropriate end of the chain for the series of dyes you're using. Students should plot L vs. b to determine the average bond length, l , where b is the independent variable and L the dependent variable.

- b. Make the appropriate graph and determine the average carbon-carbon bond length (with uncertainty).

Students should see a carbon-carbon bond length that agrees reasonably well with literature values for a 1.5 bond order. The students can be encouraged to find values for some molecule (like butene) or for any conjugated system.

Here is sample student data.

Table 1 – Student-generated molecular orbital results

Dye	Exp. λ_{max} (nm)	Total number of delocalized e^- (N)	Number of C-C bonds (b)	Effective Orbital Length (Å) (L)
Carbo	559	7	4	11.63
Dicarbo	651	9	6	14.05
Tricarbo	755	11	8	16.57

³ Moog, R. S. Determination of Carbon-Carbon Bond Length from the Absorption Spectra of Cyanine Dyes. *J. Chem. Educ.* **1991**, 68, 506-508.

Table 2 – Student-generated Molecular Orbital Results

Dye	Exp. λ_{max} (nm)	HOMO (eV)	LUMO (eV)	ΔE (eV)	Theor. λ_{max} (nm)
Carbo	559	3.74	5.82	2.08	597
Dicarbo	651	4.14	5.94	1.80	689
Tricarbo	755	4.14	5.62	1.48	838

32. In general, λ_{max} reflects the transition between two electronic states (the HOMO—highest occupied molecular orbital—and LUMO—lowest unoccupied molecular orbital) of a molecule in which each state has significant energy due to the vibrational and rotational motion of the molecule. However, the model we use assumes that there is no vibrational and rotational motion in the molecules. It has been suggested that using a wavelength greater than λ_{max} at which the absorbance is 1/5 its maximum value is a better approximation of the HOMO-LUMO transition in these molecules.³ Add a column to your spreadsheet with $\lambda_{1/5}$, and use these values to calculate the average length of the C-C bond in your dye.

This analysis follows that of the Moog article referenced in the lab handout. Students may see somewhat better results using this technique.

33. Does the value for the typical length of a carbon-carbon bond in a conjugated chain better support the use of λ_{max} or $\lambda_{1/5}$ in this analysis?

See above.

Pre-experiment Questions: Part Four

34. For each dye, determine the number of π electrons that are in the conjugated chain bounded by the two nitrogen atoms and the number of molecular orbitals that can be occupied by the number of π electrons in the conjugated chain.
35. Sketch an energy level diagram (a ladder) for the π electrons in the longest dye. Assuming that each energy level corresponds to a molecular orbital, label the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) on your diagram. Is spacing between energy levels equidistant?
36. Use an arrow on your diagram to show the energy difference that corresponds to λ_{max} .
37. Recall that the particle-on-a-line model confines particles using a square “well” with infinitely high walls (i.e., the potential energy is zero for the length of interest and infinite outside that length). Why is it physically reasonable to consider a finite rather than an infinite square well for the π electrons?
38. Suppose the well depth is changed from infinite to a finite value (such as 10.0 eV). Predict whether λ_{max} for this finite well will be the same as, larger, or smaller than λ_{max} for the infinite well.

Any prediction at this point is acceptable. The point is to get a prediction from each student.

Experiment Part Four

Focus question: What happens if we change the particle-on-a-line model to one that is more physically reasonable?

The protocol uses a Phet to model the system using a finite square-well potential. Instructors could opt to use a different program to do this modeling.

Theory Protocol

Use an online simulation to determine the energy levels of the orbitals in the conjugated chain. The simulation model assumes the electrons are confined by walls with finite potential energy (a finite square well). Therefore, the electron could escape if its energy is sufficiently high.

- A. In a web browser go to <http://phet.colorado.edu/en/simulation/bound-states>. Click “Run Now” to open the simulation window.

Instructors should check for Java issues before class in using this and other PhET simulations. Instructors may wish to encourage students to play with the simulation before moving on to the next steps. Students should focus on the One Well tab and the Square potential well. They should scroll over the green energy levels to see that the value of the energy level and the wavefunction is provided. They can manipulate the arrows on the diagram to show that the spacing of the energy levels is dependent on both the depth of the well and the width. They can also see that the offset does not affect the spacing of the energy levels.

- B. Within the simulation window and under the ‘One Well’ tab, set the ‘Potential Well’ to Square and in ‘Configure Potential ...’ set the Offset = 0.0 eV, Height = 20.0 eV.

The height used here is the largest one that the simulation allows. It would be nice if the simulation permitted infinite walls.

- C. Set the Width to each of the lengths determined in Part III for your dye molecules. For each length, record the energies for the molecular orbitals containing the delocalized electrons for each molecule in your dye series. Also record the energy for the lowest energy molecular orbital that does not contain any delocalized electrons.

While students are asked to obtain the energy levels for all occupied molecular orbitals, they will only use the energies for the HOMO and LUMO.

Thinking About the Data: Part Four

39. Using each of the two protocols, determine the energy of the HOMO and LUMO for each dye, the value of the HOMO-LUMO energy gap, and the wavelength corresponding to that gap. Compare that wavelength to the value that you used to calculate the value of L for the dye molecule. Which protocol gives a better agreement?

Students should again be doing this work in a spreadsheet or similar program. If they only use one protocol, then the last question should just have them compare the results to the experimental values.

40. How does the width of the potential well affect the energy levels in the well? Likewise, how does the length of the conjugated chain in the dye molecules affect λ_{max} for the dye molecules?

The connection between the length of the conjugated chain and the width of the well should become evident.

41. Does each energy level in the infinite square well have a higher or lower energy than the corresponding level in the finite square well?

The finite square well should have slightly lower energies than the infinite square well.

42. Explain why the lowest energy (longest wavelength) transition that can occur in the dye is the HOMO-LUMO transition.

An energy level diagram can show that any transition from an occupied molecular orbital to an unoccupied orbital will have higher energy.

43. Describe the process occurring when light (from a spectrophotometer) interacts with the electrons in each dye molecule.

If the light has an energy that corresponds to a difference in two energy levels in the dye molecule *and* the lower energy level is occupied and the higher energy level is unoccupied (or not fully occupied), then an electron in the lower energy level may absorb the photon and transition to the higher energy level.

Post-experiment Questions

44. View the “phet” *Molecules and Light*, available at <http://phet.colorado.edu/en/simulation/molecules-and-light>.

Compare the effects of four different wavelength “regions” (microwave, infrared, visible, and ultraviolet) on the different molecules.

- a. Which molecules are affected by microwave and infrared radiation, and what is the effect? Which are affected by visible light, and what is the effect? Which are affected by ultraviolet light, and what is the effect?

Students should start to see that not all molecules absorb microwave and infrared radiation, and those that do have rotations and vibrations excited, respectively. Students will also see that not all molecules absorb visible or ultraviolet. The effect of this light can be a transition to an excited electron state followed by photon emission as the molecule returns to the ground state or, in the case of ultraviolet light, bond-breaking.

- b. For the molecules that are affected by the visible light, interpret the halo that appears and what happens when the halo disappears.

The halo represents the molecule in the excited state; a new photon is emitted when the molecule returns to the ground state (indicated by the halo disappearing).

- c. Which wavelength regions correspond to the spectra you obtained, and what is the effect of those regions, if any, on the dye molecules?

In this experiment, the wavelengths are in the visible region, so the dye molecules undergo photon absorption and transition to an excited electronic state.

45. Discuss how spectroscopy may be used to determine the concentration of some molecule. What must be known about the molecule?

There is generally a need to understand the linear relationship provided by Beer's Law and how it can be applied in spectroscopy. The extinction coefficient must be known (although one could develop a calibration curve without ever calculating the extinction coefficient.)

46. In what ways, if any, does a spectrum of a dye mixture represent a *sum* of the spectra of the individual dyes? In what ways, if any, does that spectrum represent an *average* of the spectra of the individual dyes? Provide an explanation for the observed spectra based on the models developed in this experiment.

We want students to recognize that each molecule in a dilute solution behaves independently of the other molecules. That is the idea behind Beer's Law for a solution of a single molecule. The same rule applies in the mixture and that should lead to the conclusion that the spectrum of a dye mixture is the sum of the spectra of the individual dyes. Students should see that is consistent with what they observed in Part 2.

47. Three series of cyanine dyes that are readily available are the 2,2'-cyanine series, the 4,4'-cyanine series, and the thiacyanine series. Look up the structures of three or more molecules in one of these series (other than the one you've already measured). Describe the trend you expect for λ_{max} , both within the series and in comparison to the molecules studied in lab. For example, predict whether the new series will have lower or higher values of λ_{max} than the cyanine series you've already studied, and explain your prediction. If time permits and the dyes are available, measure the spectra of a second series and perform the analysis of Parts III and IV.

This question can lead to an independent project. Students may also wish to look for other series of molecules, particularly dye molecules that could be explored further. Advanced students may wish to explore quantum dots.⁴

48. While the 2,2'-cyanine and 4,4'-cyanine dyes are well known, there is little discussion in the literature of 3,3'-cyanine dyes. Try drawing the structure of 1,1'-diethyl-3,3'-cyanine analogous to the 2,2' and 4,4' systems. Use this structure to explain why there might be little interest in the 3,3' systems. (Hint: Make sure to avoid five bonds to carbon.) Also explain why the 2,4' systems would be of interest as dyes (and some have, in fact, been synthesized).

The 3,3' series of cyanine dyes does not appear to support a resonance structure in the polymethine chain, but the 2,4' series does.

49. Beginning with the particle-on-a-line model, derive the equation for λ_{max} in Thinking about the Data, Part 3. What are the highest-occupied and lowest-unoccupied molecular orbitals for the π -electrons in this system, and why is that information necessary to determine ΔE for the lowest energy transition?

⁴ See, for example, Laubach, T. A.; Elizondo, L. A.; McCann, P. J.; Gilani, S.; Quantum Dotting the "i" of Inquiry: A Guided Inquiry Approach to Teaching Nanotechnology. *Phys. Teach.* **2010**, *48*, 186-188.

50. If directed by your instructor, use a program like Hyperchem or Gaussian to calculate the molecular orbitals and energy levels of the dye molecules. Instructions will be provided. Identify the energy levels that correspond to the HOMO and LUMO of each dye molecule, and calculate a value of λ_{max} . Compare these results to your experimental values and the values from the particle-on-a-line model.

This question provides another approach to determining HOMO and LUMO energies based on more rigorous calculations. Analysis should proceed as in Part Four of the experiment.

51. The measured extinction coefficient for a spectral absorption is proportional to a quantity known as the transition dipole moment for the excitation of an electron from one energy level to another.

- d. Label the energy levels for the dye molecule starting with $n = 1$ for the lowest energy level, $n = 2$ for the next highest, and so on. What values of n correspond to the HOMO and LUMO for each dye?

The value of n for the HOMO is $N/2$, where N is the number of delocalized π electrons in the conjugated chain, and the value of n for the LUMO is $N/2+1$.

- e. For the particle-on-a-line model, the transition dipole moment μ_{T} for a given transition from initial level n_i to a final level n_f can be calculated from the relation⁵

$$\mu_{\text{T}} = \frac{8eL}{\pi^2} \left(\frac{n_f n_i}{(n_f^2 - n_i^2)^2} \right)$$

where e is the charge on the electron and the difference $n_f - n_i$ is an odd number.

When $n_f - n_i$ is an even number, $n_f - n_i$, and the transition is said to be forbidden.

Using complete sentences, describe what happens to μ_{T} as i) L gets larger; ii) the difference $n_f - n_i$ becomes large.

As L gets larger, the value of the transition dipole moment increases. But as the difference $n_f - n_i$ gets larger, the value of the transition dipole moment decreases rapidly.

- f. Using the HOMO-LUMO transition, calculate μ_{T} for each of the dyes and describe any trend that you observe. Compare that trend to the observed trend in the extinction coefficients of the dyes.

Students should see that the molecule with the longest chain has the largest transition dipole moment and that agrees with their observed results.

⁵ Hanson, D. M.; Harvey, E; Sweeney, B; Zielinski, T. J. Quantum States of Atoms and Molecules. American Chemical Society, Division of Chemical Education, 2005.

[https://chem.libretexts.org/Textbook_Maps/Physical_and_Theoretical_Chemistry_Textbook_Maps/Map%3A_Quantum_States_of_Atoms_and_Molecules_\(Zielinski_et_al.\)](https://chem.libretexts.org/Textbook_Maps/Physical_and_Theoretical_Chemistry_Textbook_Maps/Map%3A_Quantum_States_of_Atoms_and_Molecules_(Zielinski_et_al.)) (accessed February 9, 2017).

- g. Consider your answer to question 00. Why will the HOMO-LUMO transition be the strongest observed peak in the spectrum? (Interpret “strongest peak” as having the largest transition dipole moment of any allowed transition.)

Any other transition will have a larger value of $n_f - n_i$ and that will result in a lower transition dipole moment.

52. Spectra of cyanine dyes often show a shoulder on the high energy (low wavelength) side of the HOMO-LUMO absorption peak. If shoulders appear in any of your spectra, describe the experimental conditions that may make them more likely to appear and/or stronger. Perform a search of the literature for explanations of the existence of these shoulders in cyanine dye spectra and summarize those explanations and how they were tested experimentally. Can the experimental protocol in this lab be modified to perform such tests?

This question provides a foundation for further student research. Garland, et al. note that the shape of the absorption band for the cyanine dyes may change at higher concentrations due to dye dimerization.⁶ However, West and Pearce stress that dimerization occurs primarily in aqueous solutions.⁷ They note that Beer's Law is followed in alcoholic solutions over at least two orders of magnitude in concentration, making dimer formation an unlikely explanation for the appearance of the shoulder. They instead attribute the observed structure in the absorption band to a vibronic progression in the HOMO-LUMO transition. Further evidence that this question is suitable for undergraduates to explore is in the work of Zarow and Shin who performed an extensive study of the spectroscopy of cyanine dyes in methanol solution.⁸ They showed that the observed spectra were consistent with a vibronic progression, but not consistent with dimer formation.

⁶ Garland, C. W.; Nibler, J. W. Shoemaker, D. P. *Experiments in Physical Chemistry*, 8th ed.; McGraw-Hill: New York, 2009. pp 393-398.

⁷ West, W.; Pearce, S. The Dimeric State of Cyanine Dyes. *J. Phys. Chem.* **1965**, *69*, 1894-1903.

⁸ Zarow, A.; Shin, Y. Cyanine Dyes: Fine Structures in Their Absorption Spectra. *J. Am. Undergrad. Res.* **2008**, *8*, 7-14.